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MILITARY SPECIFICATION

BOILERS, STEAM, HIGH PRESSURE WATER TUBE, PACKAGED TYPE, (5,000 TO 50,000 POUNDS PER HOUR FIRED BY COAL, OIL, GAS OR COMBINATION OIL/GAS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers package-type, coal-fired or convertible to coal firing, natural circulation water tube, steam boilers. The boilers are completely integrated shop-assembled, self-contained units.

1.2 Classification. Boilers shall be of the following types as specified (see 6.2.1 and 6.5):

Type I - Equipped for coal and oil fuel.
Type II - Equipped for coal and gas fuel.
Type III - Equipped for oil and gas convertible to coal firing.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
GG-G-76 - Gages, Pressure and Vacuum Dial Indicating, (for Air, Steam, Oil, Water, Ammonia, and Chloro-Fluoro Hydrocarbon Gases.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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HH-B-671 - Brick, Refractory, Fire-Clay.
RR-G-661 - Grating, Metal, Bar Type (Floor Except for Naval Vessels).
TT-P-28 - Paint; Aluminum, Heat Resisting (1,200°F).
VV-F-815 - Fuel Oil, Burner.
WW-P-406 - Pipe, Steel (Seamless and Welded) (For Ordinary Use).

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-717 - Castable Mix, Refractory, High-Temperature Hydraulic Setting.
MIL-B-3180 - Boiler and Related Equipment, Packaging of.
MIL-F-3541 - Fitting, Lubrication.
MIL-R-18115 - Regulator, Boiler Feed Water, Automatic.
MIL-M-18221 - Meter, Flow Rate Indicating.
MIL-B-18796 - Burner, Single, Oil, Gas, and Gas Oil Combination, (400,000 BTU's Per Hour Over Input Capacity).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

ABMA - Lexicon Steam Generating Equipment.

(Application for copies should be addressed to the American Boiler Manufacturers Association, Suite 317, AM Building, 1500 Wilson Boulevard, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

B1.1 - Unified Screw Threads.
B2.1 - Pipe Threads (Except Dryseal).
B2.2 - Dryseal Pipe Threads.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A588 - High Strength, Low-Alloy Structural Steel with 50,000 PSI Minimum Yield Point to 4 Inches Thick.
D2156 - Test for Smoke Density in the Flue Gases from Distillate Fuels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section I, Power Boilers.
ASME Performance Test Code.
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

IC-S - Standards for Industrial Controls and Systems.
MG-1 - Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.
No. 85 - Prevention of Furnace Explosions in Fuel Oil- and Natural Gas-Fired Watertube Boiler-Furnaces With One Burner.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Ave., Boston, MA 02210.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL296 - Standard for Safety, Oil Burners.
UL726 - Standard for Oil-Fired Boilers.
UL795 - Commercial-Industrial Gas-Heating Equipment.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Description. The boilers shall be single or multiple drum, water tube, natural circulation, packaged type steam boilers. The boilers shall be provided with a furnace designed to permit dual firing or conversion to coal firing in minimum time and with minimum man-hours of plant labor. Boilers shall be equipped with single burners for oil and gas firing, mechanical draft equipment, electronic programing and flame safeguard controls, boiler limit, and fuel safety interlocks, modulating

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combustion controls, and soot blowers when coal and oil fired, and other auxiliary equipment and appurtenances specified herein. Control panels shall be of the console type, mounted integrally at the side of the boiler unless free standing panels are required. In cases where space requirements for the specified control system components preclude an orderly and accessible arrangement of components in a console type panel, free standing panels may be furnished subject to the approval of the contracting officer. Superheaters, when required by the specified design conditions, shall be installed as an integral part of the boiler. Fuel pump and heater sets, economizers, and free-standing panels, when furnished, may be packaged as separate units. Boilers shall otherwise be shop-assembled to the maximum practicable extent except for readily installed appurtenances, such as safety valves, feed water regulators, and water columns, which may be removed when required to increase shipping clearances and to prevent damage in transit. No deviations from the specified shipping requirements will be acceptable without prior written approval of the contracting officer. Nomenclature of ABMA Lexicon, Steam Generating Equipment shall apply.

* 3.2 Standard commercial product. Each boiler of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the boilers being furnished. Standard commercial product is a product, which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

* 3.3 Interchangeability. All boilers of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and accessories. All screw threads shall be in accordance with ANSI B1.1, B2.1, and B2.2 as applicable.

* 3.4 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials in the manufacturing of the boilers as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Recycled and recovered raw materials (such as glass, metals, paper, etc.) are produced directly from fresh substances, like virgin metal is made from ore. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

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* 3.5 Design. Boilers shall be designed in accordance with ASME Boiler and Pressure Vessel Code, Section I, Power Boilers and any additional requirements specified herein. Boilers shall be designed for the following conditions, as specified (see 6.2.1):

(a) Boiler design pressure.
(b) Operating pressure at steam drum or superheater outlet, as applicable.
(c) Steam temperature.
(d) Feedwater temperature.
(e) Boiler site elevation.
(f) Maximum and minimum anticipated ambient air temperatures.
(g) Maximum continuous output in pounds (lb) of steam per hour (/hr) (see 3.6.1).

* 3.6 Performance. Unless otherwise specified (see 6.2.1), performance requirements, including heat release rates, shall be based on the following fuel characteristics:

(a) Coal (percent by weight).

Vibrating grate stokers:

Size 1 inch to 3/4 inch times 0 with a maximum 40 percent through 1/4-inch screen.

Proximate analysis:

Volatile matter	30-40 percent.
Fixed carbon	40-50 percent.
Moisture	0-10 percent.
Ash	5-10 percent.

Ash softening temperature (where height of cone equals 1/2 width -2,300° Fahrenheit (F) minimum with 20 percent iron oxide (FE_2O_3) maximum in ash.

Gross calorific value 12,500 minimum British thermal units per pound (Btu/lb).

Moving grate stoker:

Size 1-1/4 inch maximum top size, 3/4-inch minimum top size with 40 percent through 1/4-inch screen.

Proximate analysis:

Volatile matter	30-40 percent.
Fixed carbon	40-50 percent.
Moisture	0-10 percent.
Ash	5-15 percent.

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Ash softening temperature where height of cone equals 1/2 width at 2,300°F minimum.

Gross calorific value 11,500 minimum Btu/lb.

(b) Fuel oil (percent by weight, as fired).

Ultimate analysis:

H_2	10.33
C	87.87
O_2	0.50
N_2	0.14
S	1.16
Btu/lb	18,400

(c) Natural gas (percent by volume, as fired).

Ultimate analysis:

Nethane (CH_4)	90.00
Ethane (C_2H_6)	5.00
Nitrogen	5.00
Btu/cubic foot (cu ft)	1002
Btu/lb	21,824

* 3.6.1 Capacity. The boiler shall produce the specified maximum continuous output in lb of steam/hr as specified (see 6.2.1) when the specified fuels in 3.6 are fired under the design conditions stipulated in 3.5.

* 3.6.2 Efficiency. The overall efficiency shall be not less than 76 percent when fired by coal, 80 percent when fired by oil, and 75 percent when fired by gas at 35 percent to full load.

* 3.6.3 Combustion smoke and control systems. When operating at loads varying from 25 percent to full load, combustion smoke for coal fired boilers shall not exceed ASTM D2156 (Shell-Bacharach) No. 6 spot except for 3 minutes each hour and for oil and gas fired boilers shall not exceed values specified in MIL-B-18796. The combustion control systems for coal, oil, and gas fired boilers shall maintain the steam pressure at the outlet connection within the tolerance limits of the set point values of the operating control as specified in MIL-B-18796.

* 3.6.4 Furnace heat input rate. When the boiler is operating at maximum output the heat input rates to the furnace shall not exceed the values specified in 3.6.4.1, 3.6.4.2, and 3.6.4.3. The combustion exit temperatures at the furnace exit during maximum output shall not exceed

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100°F below the inital deformation temperature for fusibility when coal fired with a maximum grate temperature of 600°F, 2,100°F when oil fired, and 2,150°F when gas fired.

* 3.6.4.1 Volume heat input rate. The volume heat input rate (see 6.1.2.1), for spreader stoker fired boilers shall not exceed 30,000 Btu/hr/cu ft of furnace volume. The volume heat input rate for oil and gas fired boilers shall not exceed 64,000 Btu/hr/cu ft of furnace volume. The above values are based on boilers with a total effective radiant heating surface equal to 60 percent or less of the total furnace wall area. For each percentage increase of effective radiant heating surface over 60 percent of the total furnace wall area, the heat input may be increased 200 Btu/hr/cu ft of furnace volume but not to exceed a total increase of 8,000 Btu/hr/cu ft.

* 3.6.4.2 Grate heat release rate. The grate heat release rate shall not exceed 600,000 Btu/hr per square foot (/sq ft) of grate area for spreader stoker with traveling grate or mechanical continuous discharge grate.

* 3.6.4.3 Surface heat input rate. The maximum heat input rate in Btu/hr/sq ft of effective radiant heating surface of the furnace shall not exceed the values of Table I.

TABLE I. Maximum surface heat input rates.

Rated capacity, lb of steam/hr	Btu/hr/sq ft
Less than 30,000	100,000
30,000 to 50,000	125,000

* 3.6.5 Sound level. Unless otherwise specified (see 6.2.1), the predicted sound power level shall not exceed 90 decibels (dB) on the "A" scale of a standard sound level meter, at 3-foot station at the boiler control panel, and at a height of 4.5 feet. The sound level shall drop at least 6 dB when measured at a distance of 6 feet and 4.5 feet in height.

* 3.7 Construction. Boilers shall be constructed for the specified design conditions and performance requirements in accordance with the ASME Boiler and Pressure Vessel Code, Section I, Power Boilers, and any other additional requirements specified herein. All tubes, headers, and drums shall be so assembled that the boiler and superheater, if used, can be drained dry.

* 3.7.1 Drums. Single drum boilers may be acceptable up to 20,000 lb/hr of steam of output capacity. Multiple drum boilers are required above 20,000 lb/hr. Drums shall be fusion welded, stress-relieved, X-rayed, and complete with all necessary drum connections, manhole openings with

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ass. Metals. ments. Recycled and recovered raw materials (such as

cover and yoke nozzles, couplings, internal fittings, dry pipe, steam baffles, and other fittings as required for safe, efficient, reliable operation, and for easy cleaning and maintenance. Drum internals shall be such as to insure that the moisture content of the steam leaving the drum will not exceed 0.5 percent. For boilers operating at 125 pounds per square inch gage (psig), and above no part of a boiler drum or header surface shall be directly exposed to furnace radiation unless it is adequately protected by a water screen in such a manner that no part of the exposed drum or header surface shall be more than 2 inches from the nearest water tube.

* 3.7.2 Steam drums. In multiple drum boiler the steam drums shall be not less than 30 inches in diameter. Steam drums less than 12 feet long shall be furnished with at least one manhole; steam drums 12 feet long and over shall be furnished with a manhole at each end.

3.7.3 Water drums and headers. The mud drum or lower drum shall not be less than 24 inches in diameter and shall be furnished with a manhole at each end. When furnished, multiple lower headers shall be not less than 10 inches in diameter and have handhole accesses sufficient in size and number to permit inspection and plugging of tubes and the cleaning and rolling of tubes into the drums by means of commercially available equipment. Inspection ports shall be provided in the casing to permit inspection of lower header handhole plates while the boiler is firing. Headers for single drum boilers shall provide means for cleaning tubes.

* 3.7.4 Tubes. Tubes shall either be electric-resistance welded or seamless steel and of sufficient size to provide for free natural circulation in the proper direction at all loads. Tube bends shall be formed in a manner to permit cleaning of the internal surfaces of tubes with conventional tube-cleaning tools. Unless otherwise specified (see 6.2.1), boiler and furnace tubes shall be not less than 2 inches in outside diameter (od); other tubes shall be not less than 1-1/2 inches in od. Tubes for single drum boilers shall be inclined from the horizontal to increase natural circulation and to minimize thermal shock.

* 3.7.5 Baffles. Baffles shall be arranged, as required, to diffuse the gas properly over heating surfaces in order to obtain maximum heat absorption. The baffles shall be held securely in place without being affected by or interfering with the free expansion or contraction of the boiler. The baffles shall be constructed of refractory not less than 2 inches thick, or of steel not less than 1/4-inch thick, or of heat-resistant alloy not less than 1/8-inch thick. Tangent-tube baffles may be used in lieu of refractory or metal-plate baffles. Baffles shall be capable of withstanding the temperature under maximum load conditions.

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* 3.7.6 Extended surfaces. When used, extended surfaces for tubes shall be continuous-welded to tubes in a manner that insures permanent high thermal conductivity.

* 3.7.7 Combustion chamber. The combustion setting, walls, and floor shall be constructed in accordance with the standard commercial design utilized by the manufacturer and shall permit free expansion of the boiler without undue stress on any part of the boiler setting. The furnace wall shall be watercooled-type construction to the grate line for side and rear walls. The furnace wall may contain, in addition to the water tubes and refractory, a sealed inner casing. Water tubes shall be so located and spaced that replacement of a single damaged tube by castable refractory to prevent a protracted shutdown will not result in serious overheating. Supplemental air-cooling may be furnished. Water wall headers and supply tubes shall be protected from the products of combustion or located below the grate line. Refractory behind the water walls shall be capable of withstanding a minimum temperature of 2,000°F. Firebrick shall be as specified in 3.9, and all other materials in the furnace area shall be of the quality and thickness used in standard commercial practice for this type of boiler. Furnaces shall be of such size and shape that, while firing the specified fuel, no flame shall impinge on the tubes. For boilers having the listed rated capacities, in lb of steam/hr, furnaces shall not exceed the shape factors in Table II. The shape factor shall equal the furnace volume divided by the product of the width and height of the furnace. Furnace volume shall be as defined in 6.1.2.1. Furnace width and height dimensions shall be based on the same end dimensions as used when volume is calculated. The calculations shall be made as follows:

$$\text{Shape factor} = \frac{\text{volume}}{\text{width times height}}$$

TABLE II. Shape factors.

Rated capacity lb/hr	Shape factor	Rated capacity lb/hr	Shape factor
5,000	6.5	40,000	17.5
7,500	7.0	50,000	21.0
10,000	8.0		
20,000	12.0		
30,000	15.0		

* 3.7.8 Cinder hoppers. Cinder hoppers, required when the stoker is specified, shall have sufficient slope from horizontal to prevent ash from sticking and be located for pick up of ash for disposal (see 3.7.13). Hoppers shall be protected against erosion, be refractory tile or stainless steel cladding, and have jacketed insulation to protect personnel from contact with hot surfaces.

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* 3.7.9 Bridge wall. The bridge wall, when required with the stoker, shall be water cooled and be of adequate height to insure against delivery of raw coal to the cinder or soot hopper located beneath the slag screen.

* 3.7.10 Overfire turbulence. Stoker overfire turbulence shall be supplied and have a capacity to provide 10 percent of the combustion air at 28 inches water gage. The system shall be provided with operating air by a single, low-volume, high-pressure fan driven by an electric motor with dripproof enclosure and sealed windings. Nozzles for system shall be equipped with manometer connection and adjustable dampers fitted with locking devices and position indicators. Nozzles shall be designed, arranged, and adjusted to provide maximum combustion efficiency, furnace turbulence, and trouble-free operation. There shall be a manometer connection and a permanent manometer immediately downstream from the main air supply damper. A suitable portable manometer shall also be provided. Manufacturer's written instructions which indicate optimum pressures at all manometer connections shall be documented in an instruction manual (see 6.3).

* 3.7.11 Access doors. The boilers shall be equipped with a refactory-lined access door for internal inspection and repairing of the furnace fireside. Access to the furnace may be through the burner opening provided that the burner is of the hinged, swing-open type, having a minimum 15-inch diameter unrestricted opening. Cleanout doors for single drum boilers shall be of sufficient size and number to completely clean the external surfaces of the tubes. Forced draft fired furnaces shall have gas-tight access doors.

* 3.7.12 Observation ports. The boilers shall be equipped with observation ports fitted either with safety shields to maintain closure unless held open by the operator or with air-cooled, heat-resistant, glass closures. Ports shall permit observation of the entire furnace interior including the flippers, feeders, nozzles, and burner when used. Forced draft fired furnaces shall have gas-tight observation ports.

* 3.7.13 Automatic ash removal. When specified (see 6.2.1), ash shall be removed from the boiler setting to ash collection point by a screw or drag-link-chain conveyor. The conveyor shall operate automatically in conjunction with the coal feed mechanism and shall be adjustable to suit the ash content of the coal and various plant conditions.

* 3.8 Casing. All exterior surfaces of the boiler, excluding doors, vents, and appurtenances shall be insulated and covered with a sheet steel jacket of not less than 0.1345 inch (US revised standard gage No. 10) thickness and with materials specified in 3.9. The jacket shall have structural steel vertical struts on not more than 27-inch centers, or a combination of vertical and diagonal struts which provide equivalent

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reinforcement. When boilers are provided with double casings, the pressure casing shall have not less than 0.1345 inch (US revised standard gage No. 10) thickness, while the outer casing may be lighter gage suitable for the service. Except for the heads, the exposed section of the steam drum shall be provided with adequate insulation and a removable sheet steel jacket. Openings of suitable size and number shall be provided for inspection of the convection tube bank and attachment of stoker and burner. With a surface wind velocity of 2 feet per second, the average surface temperature of the casing shall not exceed 170°F while the boiler is operated at full rated load in an ambient temperature of 100°F. When the combustion chambers and flue-gas passages of a boiler are normally operated under pressure, the casing, when tested in accordance with 4.4.2.2 shall be capable of holding a pressure equal to 1-1/2 times the predicted maximum furnace operating pressure. When specified (see 6.2.1), the casing shall be weatherproof.

* 3.9 Refractory and insulation. The boiler refractories and insulation shall be constructed in accordance with commercial practice and shall be so installed that it permits expansion of the boiler without placing undue stress on the boiler or on the refractories or insulation. Firebrick shall meet the requirements of HH-B-671 for superduty. Castable refractory shall meet the minimum requirements of MIL-C-717. The boiler flue gas-exit duct shall be insulated between the rear boiler wall and the dust-and-fly ash collector. The steel inner-pressure casing adjacent to the tubes shall be backed by insulating blocks of millboard suitable for not less than 1,200°F continuous service.

* 3.10 Superheater. When required by the specified operating pressure and steam temperature, a superheater shall be provided. The boiler shall be sequenced for control sequence II or control sequence IV operation of MIL-B-18796 when firing oil or gas fuel. When firing, coal means shall be provided to limit maximum fuel feed and air flow to prevent heat damage to superheater at start up and at start up from banked fire. The superheater shall be furnished as a integral part of the boiler. Superheaters shall be of the manufacturer's standard design with materials capable of withstanding the temperatures and pressures prevailing under maximum load conditions. Superheater pressure drop and surface proportioning shall be such as to deliver steam of the specified pressure and temperature under 25, 50, and 75 percent and full load steaming conditions with minimum temperature variations of 10 percent for each percentage after steady state steam flow conditions have been established. The superheater shall be equipped with drain vent (see 3.7), thermocouple connections, outlet nozzle, and safety valve nozzles.

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* 3.11 Coal stoker system.

3.11.1 Mechanical stoker. Unless otherwise specified (see 6.2.1), the stoker shall be an inclined vibrating grate or chain grate. The stoker assembly shall include a coal hopper feeder, front wall, overfire air nozzles (see 3.7.10), grate, tuyeres ducts, and piping with drives and means for automatic control of combustion. The stoker assembly shall be factory assembled and shall require only bolting to the boiler, and service and control connections, at the boiler manufacturer's shop for type I and type II and at site for type III as a conversion to coal firing. The assembly shall bolt to the boiler front in such a manner as to be commercially tight. The stoker shall be capable of maintaining an efficient fire, with CO_2 content of flue gas not less than 13.5 percent at rated load and not less than 10 percent at 1/4 rated load. The stoker rating shall be not less than that required to produce the specified boiler capacity (see 3.6.1).

* 3.11.2 Hopper. The coal hopper shall be constructed of steel plate having minimum thickness of 3/16 inch, shall conform to ASTM A588, and shall operate without bridging of fuel. Coal capacity shall be as specified (see 6.2.1).

3.11.3 Feed-gate. The feed-gate shall provide means to deposit coal upon the grate and distribute it uniformally. The feed-gate shall be cooled by natural air circulation on both sides. The gate shall be protected from abrasion by replaceable cast iron shoes. The gate shall be provided with an apron to prevent coal from accumulating on the gate. The gate shall be easily adjustable and provided with an indicator to show thickness of fuel bed.

* 3.11.4 Front wall. The front wall shall be of proper size to match the boiler and shall bolt to the boiler to form a gas-tight seal. The front wall shall be constructed of cast iron or fabricated steel and have one or more firing doors and one or more cleanout doors. The firing and cleanout doors shall be of sufficient size and located to facilitate removal of clinkers, to permit shoveling of clinkers from ash hopper, and to provide access to furnace for inspection. For multiple fuel boilers, the auxiliary burner shall be located above the stoker front wall or the auxiliary burner shall be attached to a front wall that is interchangeable with the stoker front wall.

* 3.11.5 Grate. The grate shall be of the mechanical oscillating, vibrating, dumping, or traveling type with mechanical front or rear ash discharge as specified (see 6.2.1). The grate surface shall be alloy cast iron with replaceable grate bars. Air shall be admitted to the fuel through tuyeres or nozzles designed for proper combustion, maximum grate cooling effect, and minimum resistance to air flow. The grate, unless water cooled, shall be wired for and equipped with two thermocouples so located that fuel bed condition can be indicated by temperature indicating meters.

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* 3.11.5.1 Grate drive. The grate shall be driven by electric motor through mechanical or hydraulic mechanism. The drive shall be adjustable to permit balancing of the ash discharge with the firing rate and shall be controlled by the combustion control.

3.11.6 Stoker controls. Stoker controls shall conform to the applicable requirements of 3.12. The combustion control system shall maintain furnace air/fuel input in accordance with demand within the accuracy of MIL-B-18796 and within the limits required for continuous combustion and stable flame. The system requirements are as follows:

(a) All controls shall be interlocked with a manual fuel changeover switch to prevent firing of two fuels simultaneously.
(b) Furnace input shall be maintained in accordance with a suitable energy demand under all credible conditions.
(c) For start-up conditions, means shall be provided to maintain a constant volumetric total air flow (not less than 25 percent) while fuel is controlled to satisfy start-up rates and within the established limits to be compatible with air flow. Means shall also be provided and suitable operating procedures shall be established to preclude the possibility of an unsafe fuel/air ratio condition.
(d) Means shall be provided to assure adequate primary air for the required fuel input.
(e) Means shall be provided for firebank control during periods of no load and during a week end. The firebank control shall maintain steam pressure at preset limits.
(f) The coal feed flow may be taken as the rate of feed.
(g) The air/fuel mixture shall be maintained within safe limits as established by test under any boiler output conditions within the controllable operating range of the system.
(h) Provide means for setting minimum and maximum limits on fuel and air control systems to prevent these systems from providing fuel and air flows beyond the stable flame limits of the fuel burning system. These minimum and maximum limits shall be defined by the stoker manufacturer and verified by operating tests (see 4.4.4).
(i) When changing the rate of furnace input, the air flow and fuel flow shall be changed simultaneously at the proper rates to maintain safe air/fuel ratio during and after the changes. When specified (see 6.2.1), provisions shall be made for air lead and lag of fuel during changes in firing rate.
(j) Means shall be provided to permit manual control of the firing rate, when desired. However, the system shall be so designed that it shall be impossible to put fuel control on manual while the air control remains on automatic or vise versa.

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(k) On balanced draft furnaces, maintain furnace draft at the desired set point in the combustion chamber within negative 0.1-inch water gage.
(l) Means shall be provided to prevent the control system from demanding a fuel-rich mixture. Control action toward more fuel and less air shall be blocked when air/fuel ratio decreases below preset value.
(m) Means shall be provided to cut back fuel to available air. If the air/fuel ratio should decrease below an appropriate preset value, control action toward less fuel and more air shall be taken.
(n) The combustion control system shall be interlocked with the grate drive to balance the ash discharge with the firing rate.
(o) Provide means to permit as much on-line maintenance of combustion control equipment as possible.
(p) Provide a means for calibrations and check testing of combustion control and associated safeguard equipment.

3.12 Burner systems. Boilers shall be furnished with factory-installed burners, ignition and combustion systems, and associated fuel systems. Unless otherwise specified (see 6.2.1), burners and fuel systems for type I boilers shall be designed for firing oil conforming to the requirements of VV-F-815, grade 6. Supply temperature shall be as specified (see 6.2.1). Oil burners shall be of: (1) the pressure-atomizing type with return-flow or variable-nozzle design; (2) air atomizing type with compressor furnished; or (3) steam atomizing (unless steam atomizing shall not be acceptable) as specified (see 6.2.1). Burners and fuel systems for type II boilers shall be designed to fire gas. The gas shall have the following characteristics as specified (see 6.2.1):

(a) Type.
(b) Heating valves.
(c) Density.
(d) Supply pressure.

Burners and fuel systems for type III boilers shall be designed for separate firing of either of the specified oil and gas fuels. Burner turndown expressed as a percentage of full-rated capacity shall be as specified (see 6.1(c) and 6.2.1). Burner control sequence shall be as specified (see 6.1(b) and 6.2.1). Burners, fuel system, and ignition systems shall in other respects conform to the applicable requirements of MIL-B-18796 with options as specified (see 6.2.1).

3.12.1 Combustion control system. Maximum use of common components for the combustion control system and boiler limit interlocks shall be made to control both burner and stoker firing. Where used, the common

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3.13 Interchangeability. All boilers of the same classification

components shall be designed and connected by transfer switch and interlocked to prevent firing more than one fuel at a time and to deactivate specialized controls not in service. Unless otherwise specified (see 6.2.1), the combustion control system may be electrically, electronically, pneumatically, or hydraulically operated. Boiler control systems shall at least meet the applicable requirements of MIL-B-18796 and NFPA No. 85.

3.12.2 Fuel oil pump. Unless otherwise specified (see 6.2.1), type I and type III boilers shall be furnished with a fuel oil pump conforming to the requirements of MIL-B-18796.

3.12.3 Fuel oil heaters. Unless otherwise specified (see 6.2.1), type I and type III boilers shall be furnished with fuel oil heaters conforming to the requirements of MIL-B-18796. When both fuel oil pumps and heaters are furnished, they may be mounted as an integral pump and heater set. When specified (see 6.2.1), duplex pump and heater sets shall be furnished.

* 3.13 Boiler trim. Unless otherwise specified (see 6.2.1), boiler fittings, blowdown valves, blow-off piping, feed piping, water column, pressure gages, feedwater-level regulator, completely piped with bypass, feed-stop valves, check valves, safety valves, and all other boiler appurtenances shall be furnished and shall comply with the applicable requirements of ASME Boiler and Pressure Vessel Code, Section I, Power Boilers. The water column shall be completely piped up and shall include a column water gage, test valve, blowdown, three try-cocks or dual water gages, operating chains, and low water alarm. To prevent low-water burne operation, the boiler shall have two low-water cutoffs as required in MIL-B-18796. The boiler shall be provided with a continuous blowdown connection from an internal pipe running the length of the steam drum at the point of the highest concentration of dissolved solids. A chemical feed connection with connected internal distributing pipe shall be provided. The steam main shall be provided with a calorimeter connection and the feedwater line shall be fitted with a thermometer well. A stop gate valve in main steam line between blow valve and main steam header shall be provided. A flanged nozzle shall be provided between the stop gate and blow-off valve. The size of the nozzle shall be adequate to permit release of maximum capacity of boiler to atmosphere when testing the boiler. Flanged nozzle shall be equipped with blind flange and gasket.

* 3.13.1 Feedwater regulator. Each boiler shall be furnished with a water regulator conforming to the requirements of MIL-R-18115. Unless otherwise specified (see 6.2.1), regulators shall be type I, class 2. Type I regulators shall be capable of maintaining the boiler water level under all conditions, within ±1 inch of the high water level and of the low water level set points with distances between high and low water set

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points prescribed by the manufacturer. The regulator shall be provided with a bypass around the control valve. The control valve shall be equipped with a handwheel or knob for manual control and shall show indication of position.

* 3.14 Soot blowers. When coal firing or grade 6 fuel oil is required, soot blowers shall be furnished integrally with all boilers except single drum boilers with horizontal inclined tubes. Otherwise only appurtenances for installation of soot blowers shall be furnished. Soot blowers shall be steam operated, valve-in-head type, and shall be furnished complete with wall sleeves, clamps, hangers, supports, operating chains, and other appurtenances required for a complete installation. The blower elements shall be so arranged that all parts of the heating surfaces shall be cleaned of any soot deposits when rotated by manual or automatic means. Elements shall be of such length, diameter, and total nozzle area that for the operating pressure involved, there will be no significant difference in the cleaning effect between the nozzle nearest the inlet, and those farthest from the inlet of the element. Soot blower piping shall include a valve chain operated from the floor, in addition to an auxiliary steam-stop valve, and drain connection.

* 3.15 Sample cooler. Sample cooler shall be provided to reduce to 200°F or less the temperature of the water sample taken from the boiler.

* 3.16 Control panel. A control panel shall be furnished that incorporates both stoker and burner controls. The controls including operating switches, indicating lights, gages, alarms, motor starters, fuses and circuit elements of the control systems shall be mounted on the panel. The panel and components shall meet the requirements of the control panel in MIL-B-18796. Unless otherwise specified (see 6.2.1), the control panel shall be on the left side of the boiler when viewed from the front. When specified in the contract (see 6.2.1), a remote free-standing control panel shall be furnished and include construction drawings and wiring diagrams, (see 6.3). Unless otherwise specified (see 6.2.1), in addition to the above control components, the following instruments shall be mounted on the panel.

* 3.16.1 Boiler meter. A boiler meter for indicating and recording; steam, totaling steam-flow, air-flow, and flue gas temperature shall be furnished. The flow meter shall conform to the requirements of MIL-M-18221 and shall also measure and record flue-gas temperature. The temperature measuring element shall be equipped with an averaging type resistance or thermocouple sensing device. The boiler meter shall record steam-flow, relative air-flow, and flue gas temperature on one chart. One year supply of charts and ink shall be provided.

* 3.16.2 Draft gage. A multiple-point, semi-flush, draft gage designed to indicate windbox, furnace, boiler, economizer, and air heater. When used, outlet pressures shall be provided.

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* 3.16.3 O_2 indicator. O_2 indicator in the flue gas shall be provided as required in MIL-B-18796 for oil and gas firing. Unless otherwise specified (see 6.2.1), coal fired boilers shall be provided with a portable O_2 analizer.

* 3.16.4 CO_2 indicator. A flue gas CO_2 indicator shall be provided.

* 3.16.5 Pressure gages. An 8-1/2 inch steam gage, a 4-1/2 inch oil pressure gage, and a 4-1/2 inch atomizing steam gage (when a steam atomizing burner is used) shall be provided. Gages shall conform to GG-G-76.

* 3.16.6 Thermometer. A 4-1/2 inch dial type thermometer shall be provided to indicate the temperature of fuel oil at the burner and when specified (see 6.2.1), the temperatures of superheated steam and the air preheater gas outlet.

* 3.17 Draft fans. Forced and induced draft fans shall be furnished for the stoker and burner for a balanced draft system. The forced draft fan and duct shall be suitable for connecting to stoker and burner combustion air system and air heater (if provided). The induced draft fan and housing shall be connected to the boiler stack. The fans shall be electric-motor-driven for direct or V-belt drive, as specified (see 6.2.1). Unless otherwise specified (see 6.2.1), bearings for fans shall meet the requirements of FF-B-171 for an average life of 100,000 hours. The static pressure of the forced draft fan at the outlet shall be as specified (see 6.2.1), plus the maximum pressure drop (burner or stoker) to give a positive or a balanced (0) pressure as specified (see 6.2.1), at the inlet of the furnace. The inlet and outlet static pressure required of the induced draft fan shall be as specified (see 6.2.1). The induced draft fan shall be designed for handling hot combustion flue gases at a maximum temperature, as specified (see 6.2.1). The fans shall be designed and sized to conform to the draft fans of MIL-B-18796 when firing the specified fuels herein. Fan controls and interlocks shall be as specified in MIL-B-18796. Manual control switches shall be provided to permit purging prior to cold start.

* 3.18 Economizer. When specified (see 6.2.1), a separately packaged economizer shall be provided for boilers that operate at pressures over 250 psig. The economizer shall be complete with prefabricated interconnecting breeching for attachment to the boiler flue outlet and to the feed piping between the economizer and the boiler drum, and shall require minimum field assembly and erection to the boiler. Economizers shall be of the manufacturer's standard design for the operating conditions specified. Materials used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions. Soot blower(s) capable of cleaning all surfaces of soot deposits shall be provided on economizers furnished.

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* lion of refractory or metal plate baffles. Baffles shall be capable o

* 3.19 Air heater. When specified (see 6.2.1), a separately packaged air heater shall be furnished for boilers that operate at pressures over 250 psig. The air heater shall have means for bypassing air during part-load operation to maintain furnace gas outlet temperature to reduce condensation and corrosion. The unit shall be complete with soot blower and for the recuperative type shall have means for washing corrosive salts from surfaces exposed to furnace gases. Prefabricated breeching shall be provided for connection to the economizer, when specified, otherwise to dust collector. The ducts shall be prefabricated for connection to the forced draft fan and the furnace. Field fabrication shall not be permitted. The prefabricated packaged air heater shall require minimum field assembly and erection.

* 3.20 Fly ash and soot collector. A fly ash and soot collector shall be provided except when type III boiler is furnished. The collector shall be either the mechanical, centrifugal, precipitator type consisting of single large unit, or a multiplicity of individual small diameter units operating in parallel. The collector shall be mounted with tne induced draft fan and fan drive on a rigid frame of structural steel. Breechings fabricated of steel plate shall be provided between the boiler and the collector and between the collector and the fan inlet. Provisions shall be made for expansion and contraction and for a damper between the collector and fan. The hopper shall be fitted with gas-tight inspection and cleanout openings and a bin-level indicator of approved design. The hopper bottom shall be fitted as necessary to insure proper connection and discharge to the fly ash injection or the ash-removal system. All components of the fly ash collector shall be fabricated of steel plates not less than 0.1875-inch thick. Parts subject to impingement of dust-laden flue gas shall be fabricated of low-alloy, low-carbon steel plates conforming to the requirements of ASTM A588. All seams shall be permanently welded and the unit shall be adequately reinforced with structural steel. The draft loss and efficiency, at rated capacity and 50 percent rated load based on fly ash having a true specific gravity of 2.0, shall be as required to meet emission standards, as specified (see 6.2.1). A test report on particle size, distribution, efficiencies, and total percent collected shall be available when specified in the contract (see 6.3).

* 3.21 Transition piece. When specified (see 6.2.1), a transition or junction piece shall be furnished to permit adapting from the boiler outlet to stacks or breeching. Transition pieces shall be designed for vertical or horizontal discharge, as specified (see 6.2.1).

* 3.22 Stack. When specified (see 6.2.1), a stack shall be provided and suitably supported. The stack shall be the last element of the combustion flue gas discharge. The stack shall be constructed of hot-rolled steel sheets not less than 3/16 inch in thickness. The stack shall be suitable for the discharge of the combustion gases for the fuels specified herein and at all loads. The stack shall be designed for the installation, as specified (see 6.2.1).

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* 3.23 Grating. When specified (see 6.2.1), steel grating shall be nonslip rectangular bar type steel conforming to RR-G-661. Grating shall be free from warps or camber and shall bear evenly upon the supporting members. All openings and ends shall be banded with bars of equal size. The finished grating shall be hot-dipped zinc coated after fabrication. Open type nonslip steel grating shall be the standard commercial product of an established manufacturer. Catwalks shall be provided as necessary to allow access to all locations requiring servicing or maintenance.

* 3.24 Pipe railing and posts. When specified (see 6.2.1), pipe railing and posts shall be fabricated of 1-1/2 inch diameter standard mild steel pipe conforming to WW-P-406, type I, classes A and B. Joints in pipe railing shall be welded and the welds ground smooth. Bottoms of posts shall be fabricated in accordance with the details indicated for the particular locations. Pipe railings and posts shall be installed plumb and true to line.

* 3.25 Electrical wiring and equipment.

* 3.25.1 Electric motors. Electric motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG-1. Motors shall be general purpose, continuous duty, ball-bearing, induction motors. The motors shall be designed for operation on a power supply having the specified characteristics (see 6.2.1). Unless otherwise specified (see 6.2.1), motors shall be furnished with sealed dripproof enclosures and windings.

* 3.25.2 Motor starters. Unless otherwise specified (see 6.2.1), motors shall be provided with magnetic across-the-line starters with overload and under-voltage protection and manual-off-automatic control stations. Magnetic starters shall conform to the applicable requirements of NEMA IC-S and shall have NEMA type 12 enclosures.

* 3.25.3 Wiring and circuit safety devices. All electrical wiring and circuit safety devices shall conform to NFPA No. 70. All control circuits shall be two pole, three wire with ground, not to exceed 120 volts nominal, and be protected with suitable fuses or circuit breakers.

* 3.25.4 Circuit breaker. A separate circuit breaker shall be provided for the steam generator.

* 3.26 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173 except that:

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(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.27 Electromagnetic compatibility. When specified (see 6.2.1), the steam generator shall be designed and equipped for electromagnetic compatibility in accordance with class IIB equipment of MIL-STD-461 (see 4.5).

* 3.28 Treatment and painting. All exterior surfaces of the boiler, before application of the insulation, shall be cleaned by commercial blast. All interior surfaces of the sheet steel jacket, before assembly, shall be cleaned by commercial brush-off blast, and finished with one coat of aluminum heat-resisting paint conforming to TT-P-28. The paint shall be applied directly to clean bare metal surfaces and shall attain a minimum dry film thickness of 1 mil. After assembly, all exposed surfaces of the equipment normally painted in commercial practice shall be cleaned as specified herein and finished with two coats of the aluminum heat-resisting paint applied directly to the clean metal surfaces, each coat applied to minimum dry film thickness of 1 mil. Component parts such as motors, etc., procured factory finished, need not be repainted. Mechanical cleaning need not be performed on the sheet steel jacket with straw colored mill scale, provided that surfaces are free of all loose mill scale and rust. However, the surfaces shall be cleaned as necessary to remove all grease and other foreign matter.

* 3.29 Lubrication. All moving parts of the equipment normally requiring lubrication shall be provided with suitable means for such lubrication.

* 3.29.1 Lubricants. The equipment shall be designed to operate efficiently and satisfactorily when lubricated with standard Military lubricants (see 6.4).

* 3.29.2 Lubrication fittings. Lubrication fittings shall conform to types I, II, and III of MIL-F-3541 and be located in accessible, protected positions. A bright red circle shall be painted around each lubricating point.

* 3.29.3 Caution plates. When the use of high-pressure lubricating equipment, 1,000 pounds per square inch and higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

* 3.29.4 Initial lubrication service. The equipment shall be serviced before delivery to the Government with winter grades of standard Military lubricants designated for use in the temperature range of 0°F to +32°F. After the lubrication service, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of lubricants used.

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* 3.30 Lifting attachments. When specified (see 6.2.1), suitable lifting attachments shall be provided to enable the equipment to be lifted in its normal position. The attachments shall be so located that, when hoisted, adequate clearance will exist between lifting slings and all exterior parts of the equipment. When two or more slings are used, each attachment shall be of sufficient capacity to carry at least 2-1/2 times the weight being lifted. The attachments for multiple slings shall be so designed that when possible the attached slings will converge at a point not to exceed 19 feet above the lowest extremity of the load. Lifting attachments shall be positioned in such direction that the lifting strain will be in line with the longitudinal axis (peripheral plane) of the eye of the attachment. The eye of each lifting attachment shall be not less than 3 inches in diameter. The equipment members, to which pad eyes are attached, shall be of sufficient strength to withstand stresses in the amount and direction of pull specified for the pad eyes. Where required, suitable reinforcements shall be used to meet the requirements specified herein. A diagram showing the lifting attachments shall be inscribed on a copper- or zinc-base alloy plate securely fastened to the unit, with the lifting capacity of each attachment marked thereon. A silhouette of the equipment showing the center of gravity shall be provided.

* 3.31 Tiedown devices. When specified (see 6.2.1), each boiler shall be provided with suitable, permanently attached devices to permit tiedown of the units to the floor or deck of the transportation medium. The tiedown devices shall be of such design, location, and number as to enable the boiler to be anchored to prevent shifting or movement in any direction. The devices shall be designed to withstand thrusts of the following magnitude: forward thrust 8G (G = cargo weight of unit); rearward thrust 8G; upward thrust 2-1/4G; and sideward thrust 1-1/2G. The tiedown devices shall be so located as to permit easy attachment of cables or turnbuckles. Tiedown devices shall be indicated by a suitable information plate, by stenciling, or other suitable marking. The marking shall clearly indicate that the device is intended for tiedown of the boiler to be shipped on the carrier.

* 3.32 Transportability by rail. When transported by rail, the boiler shall withstand the impact forces encountered in shipment without damage or deformation to its components or mounted equipment and tiedown attachments. Each tiedown attachment shall withstand its proportion of the total thrust.

* 3.32.1 Association of American Railroads clearances. When specified (see 6.2.1), the overall dimensions of the boiler in reducible condition and positioned for transport shall meet the requirements of the Association of American Railroads (see Figure 1).

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* 3.32.2 Berne International. When specified (see 6.2.1), the overall dimensions of the boiler in reducible condition and positioned for transport shall meet the requirements of Berne International (see Figure 2).

* 3.33 Instruction plates. The boiler shall be equipped with hazard or warning instruction plates suitably located, describing any special hazard or warning and procedures to be followed. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.34 Instruction manual required. Manufacturer's publications shall be furnished in accordance with the contract and as specified on the DD Form 1423. The manual shall contain technical information pertinent to the description, assembly, installation, operation, trouble shooting, maintenance, disassembly, repair of the equipment and parts list. The operating manual shall include information and instructions relating to the light-off and shut-down of the unit and shall otherwise comply with operating instructions of MIL-B-18796. Each manual shall contain the basic operating maintenance and installation instructions required under UL726 and UL795, as applicable (see 6.3).

* 3.35 Identification marking. Unless otherwise specified (see 6.2.1), assemblies and parts shall be marked for identification in accordance with ASME Boiler and Pressure Vessel Code, Section I, Power Boilers. The nomenclature of the item shall be BOILER, STEAM, HIGH PRESSURE WATER TUBE, PACKAGED TYPE.

* 3.36 Workmanship and fabrication. Workmanship and fabrication shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section I, Power Boilers, and shall be of such quality as to produce boilers that meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Standards compliance. The manufacturer shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Field inspection (at site of installation) (see 4.2.2).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on all boilers offered to the Government. This inspection shall consist of the examination of 4.3 and tests of 4.4.1, 4.4.2.1, and 4.4.2.3. In addition to the above tests, the first production unit shall meet the test requirements of 4.4.2.2. Unless otherwise specified (see 6.2.1), the first production unit shall meet the test requirements of 4.4.3.

4.2.2 Field inspection. Unless otherwise specified (see 6.2.1), field inspection shall consist of the tests of 4.4.3, 4.4.4, and the right of the Government to conduct, at the site of installation, other field tests that may be deemed necessary by the procuring activity to verify compliance with the requirements of this specification. Test 4.4.3 may be omitted on the first production unit if the unit has been shop tested to 4.4.3. The contractor shall have the priviledge of representation when such field tests are conducted. Field tests other than the above listed tests of this specification shall be conducted at Government expense. Deficiencies noted by all field tests shall be corrected at the contractor's expense and any required retesting shall be at the contractor's expense. Field tests will be performed prior to putting the boiler in service and within 1 year from the date of delivery to the Government.

4.3 Examination. The boilers shall be examined to verify compliance with this specification. The examination shall be as specified in Table III. Any boiler containing one or more defects shall be rejected. Additional defects not listed in Table III shall be classified in accordance with MIL-STD-105.

TABLE III. Classification of defects.

Categories	Defects
Critical:	
01	Programing, combustion and safety control systems including flame safeguard control, boiler limit controls, and fuel safety interlocks not as specifed (see 3.1, 3.11.6, 3.12.1, and 3.12.2).
02	Number of safety valves and aggregate relieving capacity not in accordance with ASME Boiler and Pressure Vessel Code (see 3.5 and 3.13).

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TABLE III. Classification of defects - continued.

Categories	Defects
Major:	
101	Boilers not of the type or control system specified (see 1.2).
102	Boiler design, materials, and construction not in accordance with the applicable provisions of this specification and the ASME Boiler and Pressure Vessel Code, Section I, (see 3/4. 3.5, and 3.13).
103	Boiler not complete. Component parts including boiler trim, fittings, or accessories missing (see 3.7, 3.13, and 3.16).
104	Coal stoker not complete (see 3.11).
105	Burner not complete (see 3.12).
106	Feedwater regulator not furnished or as specified (see 3.13.1).
107	Soot blowers not as specified or not furnished (see 3.14).
108	Sample cooler missing or not complete (see 3.15).
109	Control panel not complete or located incorrectly (see 3.16).
110	Economizer, when furnished, not complete with specified accessories (see 3.18).
111	Air heater, when furnished, not complete with specified accessories (see 3.19).
112	Fly ash and soot collector not complete with specified accessories (see 3.20).
113	Wiring not in accordance with NFPA (see 3.25.3).
114	Treatment and painting not as specified (see 3.28).
115	Workmanship and fabrication not as specified (see 3.26).
Minor:	
201	Instruction plates missing (see 3.33).
202	Publications missing (see 3.34).
203	Identification markings missing or incorrect (see 3.35).

4.3.1 Furnace heat input. The manufacturer shall have available data sheets (see 6.3), including required calculations and sketches, to indicate conformance to 3.6.4. The data sheet shall include the effective radiant heating surface, as defined in 6.1.2.2 for each of the furnace walls including roof, end wall, sidewalls, burner wall, and floor as applicable. Furnace dimensions shall also be given to indicate conformance to 3.6.4.1 as defined in 6.1.2.1. Failure of the manufacturer to demonstrate adequate conformance to 3.6.4 shall be cause for rejection of the boilers.

4.4 Tests. Each boiler shall be tested in accordance with the requirements of 4.2. Failure of any boiler to pass the following tests, as applicable, shall be rejected. Failure of any boiler to pass any

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test specified in 4.4.3 and 4.4.4 shall be cause for rejection of the lot. Certified test reports (see 6.3), from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests. Tests shall be conducted as outlined in the referenced documents as specified herein.

* 4.4.1 Fire and explosion safety tests. Tests shall be conducted as necessary to determine compliance with the applicable UL Safety Standards and system safety requirements herein. The presence of the applicable UL or the certificate of compliance (see 6.3) required for MIL-B-18796 will be accepted as evidence of compliance.

4.4.1.1 Oil-fired boilers. Oil-fired boilers shall meet the test requirements of UL726.

4.4.1.2 Oil burners. Oil burners shall meet the test requirements of UL296.

4.4.1.3 Gas-fired boilers and gas burners. Gas-fired boilers and gas burners shall meet the test requirements of UL795.

* 4.4.1.4 Stoker controls. Controls shall be tested for the fire and safety system requirements of 3.11.6.

* 4.4.2 Strength and tightness tests. The boiler(s) shall be subjected to the following strength and tightness tests.

* 4.4.2.1 Watersides including fitting and accessories. Watersides shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section I, Power Boilers. The ASME stamp may be accepted as evidence of this test.

* 4.4.2.2 Boiler casing, air casings, and ducts. Air casing and ducts, exterior to the furnace, shall be tested pneumatically at the maximum working pressure. The soap bubble or the halogen-sniff test method shall be used to verify tightness. Combustion chambers and flue-gas passages of boilers normally operated under pressure shall be tested for tightness at 1-1/2 times the predicted operating pressure in the furnace at maximum continuous output. For this test, the boiler shall be tightly sealed with a suitable means to blank-off all openings. Air shall be admitted to the boiler until the test pressure is reached. The air pressure shall be held in the boiler during the test. If in a 10-minute period the pressure drop does not exceed 5 inches water gage, the casing shall be regarded as tight and accepted. Air pressure and soap bubble tests or comparative CO_2 readings shall be used for induced draft boilers.

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* 4.4.2.3 Fuel systems. The part of the preassembled fuel oil system that is furnished integrally with the boiler shall be hydrostatically tested at 1-1/2 times the maximum operating pressure. The part of the preassembled gas system that is furnished integrally with the boiler shall be pneumatically tested at operating pressure. The soap bubble or the halogen-sniff test method shall be used to verify tightness of the gas system.

* 4.4.3 Capacity and efficiency tests. The capacity, efficiency, and steam quality at the specified capacity of the boiler shall be determined in accordance with the ASME Performance Test Code PTC 4.1. The ASME Test Form for Abbreviated Efficiency test may be used. The efficiency shall be determined by the heat loss method. Performance shall be as specified in 3.6. Methods of measuring flow of steam, water, and fuel other than weighing will be allowed providing the accuracy of the method will assure the Government the desired performance specified in 3.6 and be acceptable to the contracting officer or his authorized representatives. Test runs shall be made at the maximum capacity, the minimum capacity, and at even increments of the turndown ratios specified in 3.11.1 and 3.12 as applicable. Test reports and performance curves shall be submitted to the contracting officer (see 6.3).

* 4.4.4 Operational tests. The boiler shall be tested to demonstrate control and operational conformance to the requirements of this specification and MIL-B-18796, under varying load conditions ranging from the specified capacity to the minimum burner turndown ratio without on-off cycling. All burners shall be tested for ability to start a cold boiler in accordance with the manufacturer's instructions. Unless otherwise specified (see 6.2.1), oil burners shall be tested for ability to make a cold start from the specified oil supply temperature of 3.12. Automatic burners shall also be tested for ability to provide steam in accordance with steam demand. Boilers that exhibit: (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, (5) necessity for frequent or difficult adjustments, or excessive noise shall be rejected. Operational tests shall include the following:

* 4.4.4.1 Sequencing. The boiler shall start, operate, and stop in strict accordance with the specified operating sequence.

* 4.4.4.2 Flame safeguard. The operation of the flame safeguard control shall be verified by simulated flame and ignition failures. Burners shall be tested to simulate main flame failure and spark rejection. The trial-for-pilot-ignition, trial-for-main-flame-ignition, combustion control reaction, and valve closing times shall be verified by stop watch.

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* 4.4.4.2.1 Immunity to hot refractory. The burner shall be operated at high fire until the combustion chamber refractory reaches maximum temperature. The main fuel valve shall then be closed manually. The combustion safeguard shall de-energize the safety shutoff valves within the specified control reaction time and the valves shall close within the specified valve closing times.

* 4.4.4.2.2 Pilot intensity required. The fuel supply to the pilot flame shall be gradually reduced to the point where the combustion safeguard begins to drop out (sense "no flame") but holds in until the main fuel valve opens. At this point of reduced pilot fuel supply the pilot flame shall be capable of safely igniting the main burner. If the main fuel valve can be opened on a pilot flame of insufficient intensity to safely light the main flame, the boiler shall be rejected.

* 4.4.4.3 Burner turn-down ratio. The specified burner turn-down ratio shall be verified by firing at the minimum firing rate required by the specified turn-down ratio.

* 4.4.4.4 Boiler limit and fuel safety interlocks. Safety shutdown shall be caused by simulating interlock actuating conditions for each boiler limit and fuel safety interlock. Safety shutdowns shall occur in the specified manner.

* 4.4.4.5 Combustion controls. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the steam demand through the entire firing range required by the turn-down ratio specified for the burner and in case of automatic sequenced burners, by further varying the firing rate to require on-off cycling. The control accuracy shall be as specified in 3.11 and 3.12.

* 4.4.4.6 Safety valves. The high pressure limit switch shall be locked out or otherwise made inoperative and the boiler safety valves shall be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure shall be determined by observation and measurement to be in accordance with the ASME Boiler and Pressure Vessel Code. This test shall be made in the presence of the contracting officer or his authorized representative. The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the code.

* 4.4.4.7 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

* 4.4.5 Fans, heaters, pumps, and motors. Draft fans, fuel oil heaters, fuel pumps, and electric motors shall be tested, when necessary, to determine compliance with the referenced standards. Standard symbols and certifications (see 6.3) from the referenced organization may be

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accepted at the discretion of the contracting officer or his authorized representative. The operation of fans, fuel oil heaters, fuel pumps, and electric motors shall be closely observed for possible defects or nonconformance.

* 4.5 Electromagnetic interference tests. When electromagnetic interference control is specified (see 3.27), the first production unit shall be tested in accordance with MIL-STD-462, to verify compliance with the pertinent provisions of MIL-STD-461.

* 4.6 Packaging inspection. The preservation-packaging, packing, and marking of the boiler shall be inspected to determine conformance to the applicable requirements of Section 5 of this specification.

5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A. The boilers shall be preserved and packaged in accordance with the applicable level A requirements of MIL-B-3180.

5.1.2 Level C. The boilers shall be preserved and packaged in accordance with the supplier's standard practice.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. The boilers shall be packed in accordance with the applicable level A requirements of MIL-B-3180.

5.2.2 Level B. The boilers shall be packed in accordance with the applicable level B requirements of MIL-B-3180.

5.2.3 Level C. The boilers shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

* 6.1 Intended use. The boilers covered by this specification are intended for generation of steam to be used in processing operations, space heating, or power generating systems in land installations.

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* 6.1.1 Selection factors. The following should be considered in the selection of a boiler and the exercise of options herein:

(a) Dual firing capability with two fuels and the availability or economics of the fuel supply should determine the type of boiler.

(b) Automatic control sequences should not be specified for boilers furnished with superheaters. Superheaters should be limited to boilers with means for starting with low fire until saturated steam enters the superheater and then manually raise the combustion rate to high fire to prevent overheating. Burners operating in occupied buildings or in hazardous locations require a safety control system that includes repetitive self-checking circuits, such as process heating applications, and step-firing.

(c) Burners should be specified in accordance with MIL-B-18796 with specified options. Selection of combustion controls should be as recommended in MIL-B-18796. Most burners in MIL-B-18796, when equipped with modulating controls, will have minimum firing rates of 20 to 25 percent of the maximum firing rates. Burners having input capacities above 25 million Btu/hr may be readily obtained with minimum firing ratio of 16.7 percent of the maximum rate. Steam atomizing burners should be specified only if the boiler operating pressure exceeds 100 psig.

(d) Standard design pressures for boilers covered by this specification range from 150 to 800 psig. Superheaters can be provided to furnish steam up to 200°F above saturation temperature but with a maximum temperature limit of 700°F.

6.1.2 Definitions.

* 6.1.2.1 Furnace volume. Furnace volume shall be defined as the cubical volume above the top grate line for coal firing (or the floor for oil and gas firing when grate is not covered) up to the first plane of entry into or between the tubes. Screen tubes, when utilized, shall be considered as the first plane of entry. The space occupied by a radiant type superheater (if used) shall not be considered furnace volume.

* 6.1.2.2 Effective radiant heating surface. Effective radiant heating surface is defined as heat exchange surface, exclusive of superheater elements, which is directly exposed to the radiant heat of the flame on one side and to the water or water-steam mixture being heated on the other. Effective radiant heating surface shall be calculated on the side receiving heat and shall consist of plain or finned tubes and headers which may be bare, metal covered, or metallic-ore covered. Refractory-covered surfaces or the floor covered grate shall not be counted in the computation of the metal effective radiant heating surface. Computations shall be based on the following consideratio

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(a) The flat projected area of bare, metal covered, or metallic-ore covered tubes and headers shall be considered as effective radiant heating surface.
(b) Metal and metallic surfaces extending from tubes or headers shall be considered to have an effective radiant heating surface equal to 80 percent of their flat projected area except that the following extended surfaces shall not be considered as effective radiant heating surface:

(1) Metal blocks not integral with tubes or headers.
(2) Extended surfaces less than 1/4-inch thick.
(3) That portion of the extended surface more than one tube or header radius from the tube or header from which it extends.
(4) Extended surfaces longer than 1-1/4 inches.

(c) The flat projected areas of the portions of the first two rows of furnace exit tubes that receive radiant heat from the fire shall be considered as effective radiant heating surface.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Design conditions and capacity (see 3.5 and 3.6.1).

(1) Boiler design pressure.
(2) Operating pressure.
(3) Steam temperature.
(4) Feedwater temperature.
(5) Boiler site elevation.
(6) Maximum and minimum anticipated ambient air temperatures.
(7) Maximum continuous output in lb of steam/hr.

(d) Proximate analysis and heating value of coal and ultimate analysis and heating value of intended oil or gas fuel if different (see 3.6).
(e) When sound level shall be different (see 3.6.5).
(f) When tubes shall be of a size other than as specified (see 3.7.4).
(g) When automatic ash removal is required (see 3.7.13).
(h) When weatherproof casing is required (see 3.8).
(i) When stoker shall be of another type than as specified (see 3.11.1).
(j) Required coal capacity of hopper (see 3.11.2).

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(k) Type of grate required and when front or rear ash disposal is required (see 3.11.5).
(i) When provisions shall be made for air lead and lag of fuel during changes in firing rate (see 3.11.6(i)).
(m) Grade of fuel oil if different for type I boiler (see 3.12).
(n) Required fuel oil supply temperature for type I boiler (see 3.12).
(o) Type of oil burner required for type I boilers (see 3.12).
(p) Fuel gas characteristics for type II and type III boilers (see 3.12).

(1) Type.
(2) Heating value.
(3) Density.
(4) Supply pressure.

(q) Required burner turndown in percent of full rated capacity (see 3.12 and 6.1.1(c)).
(r) Required burner control sequence (see 3.12 and 6.1.1(b)).
(s) Required options to MIL-B-18796 (see 3.12 and 6.1.1).
(t) When a specific type of actuation (pneumatic, electric, or hydraulic) is required for combustion control system (see 3.12.1).
(u) When fuel oil pump shall not be furnished (see 3.12.2).
(v) When fuel oil heaters shall not be furnished; when duplex pump and heaters shall not be furnished (see 3.12.3).
(w) When boiler trim is different (see 3.13).
(x) When feedwater regulator is different (see 3.13.1).
(y) When location of control panel is different (see 3.16).
(z) When remote free-standing control panel with construction details is required (see 3.16).
(aa) When number and type of instruments are to be different than specified (see 3.16).
(bb) When O_2 analizer shall be different for coal fired boilers (see 3.16.3).
(cc) When thermometers are to be provided to indicate temperatures of superheated steam and air preheater gas outlet, when used (see 3.16.6).
(dd) Type of drive required for draft fans (see 3.17).
(ee) When type of bearings for fans shall be different (see 3.17).
(ff) For balanced draft system, static pressure required at the forced draft-fan outlet; the positive or balanced (0) static pressure at the inlet of the furnace; the inlet and outlet static pressure required of the induced draft fan; and the maximum temperature of the flue gas (see 3.17).
(gg) When an economizer is required (see 3.18).
(hh) When an air heater is required (see 3.19).
(ii) Required emission standards (see 3.20).
(jj) When transition piece is required and whether horizontal or vertical discharge for transition piece is required (see 3.21).

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(kk) When stack is required and design details of installation (see 3.22).
(ll) When grating is required (see 3.23).
(mm) When pipe railing and posts are required (see 3.24).
(nn) Required electrical power supply characteristics (see 3.25.1).
(oo) When motors are to be furnished with other than dripproof enclosures with sealed windings (see 3.25.1).
(pp) When motor starters are to be different (see 3.25.2).
(qq) When fungus and moisture resistance is required (see 3.26).
(rr) When electromagnetic compatibility is required for boiler (see 3.27).
(ss) When lifting attachments are required (see 3.30).
(tt) When tiedown devices are required (see 3.31).
(uu) When overall transport dimensions shall meet Association of American Railroad clearances (see 3.32.1).
(vv) When overall transport dimensions shall meet Berne International (clearance diagram) (see 3.32.2).
(ww) When identification marking shall be other than as specified (see 3.35).
(xx) When test requirements of 4.4.3 shall be different for the first production unit (see 4.2.1).
(yy) When test requirements for field inspection shall be other than specified (see 4.2.2).
(zz) When oil burners shall not be tested for their cold-start ability at oil supply temperature (see 4.4.4).
(aaa) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DD 1664
3.7.10, 3.34	Instruction manuals	DI-M-24010A
3.16	Drawings	DI-E-7014A
3.20, 4.4.3	Test report	DI-T-2072
4.3.1	Design data and calculations	DI-R-24040A
4.4.1, 4.4.5	Certificate of compliance	DI-E-2121

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(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 Standard lubricants. When requested by the contractor, the contracting officer should obtain, through established channels, a list of standard Military lubricants applicable to the equipment covered by this specification. This information is contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL.

6.5 Supersession data. This document differs from the superseded specification in the following classifications:

MIL-B-23170(DOCKS) (old)	MIL-B-23170A (new)
Type I - Equipped for coal fuel.	Type I - Equipped for coal and oil fuel.
Type II - Equipped for oil fuel.	Type II - Equipped for coal and gas fuel.
Type III - Equipped for gas fuel.	Type III - Equipped for oil and gas fuel convertible to coal firing.
Class 1	Deleted
Class 2	Deleted
Class 3	Deleted
Class 4	Deleted
Class 5	Deleted
Class 6	Deleted
Capacity	Deleted see paragraph 3.6.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

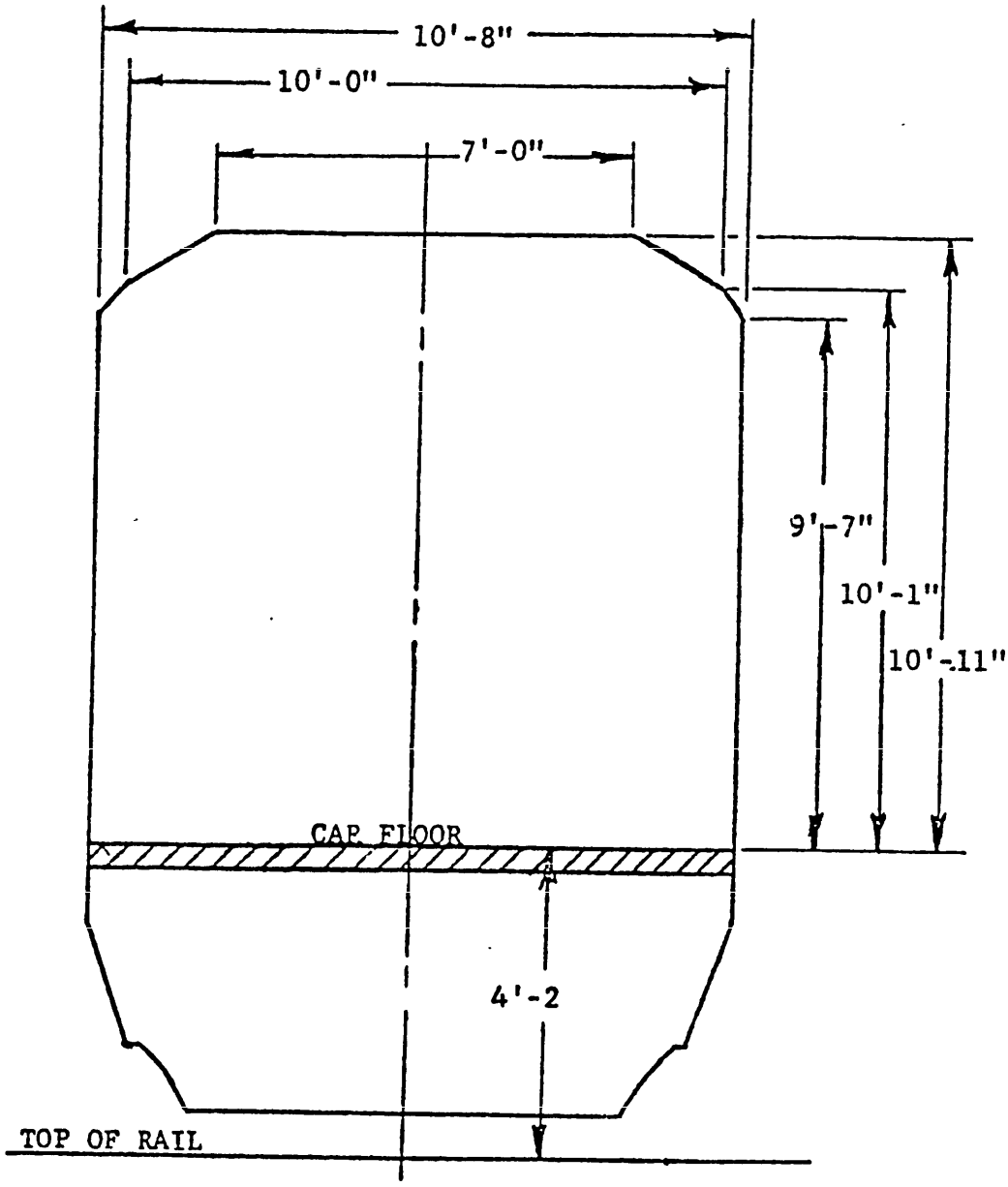
Navy - YD

Project No. 4410-0039

User activity:

Army - CE

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NOTE: DIAGRAM COVERS CARS OF DIMENSIONS UP TO:

Inside Length----------50'-6"
Coupled Length----------54'-6½"
Center Line to Center Line-------41'-3"

FIGURE 1: Outline Diagram of Approved Limited Clearances of ASSOCIATION OF AMERICAN RAILROADS.

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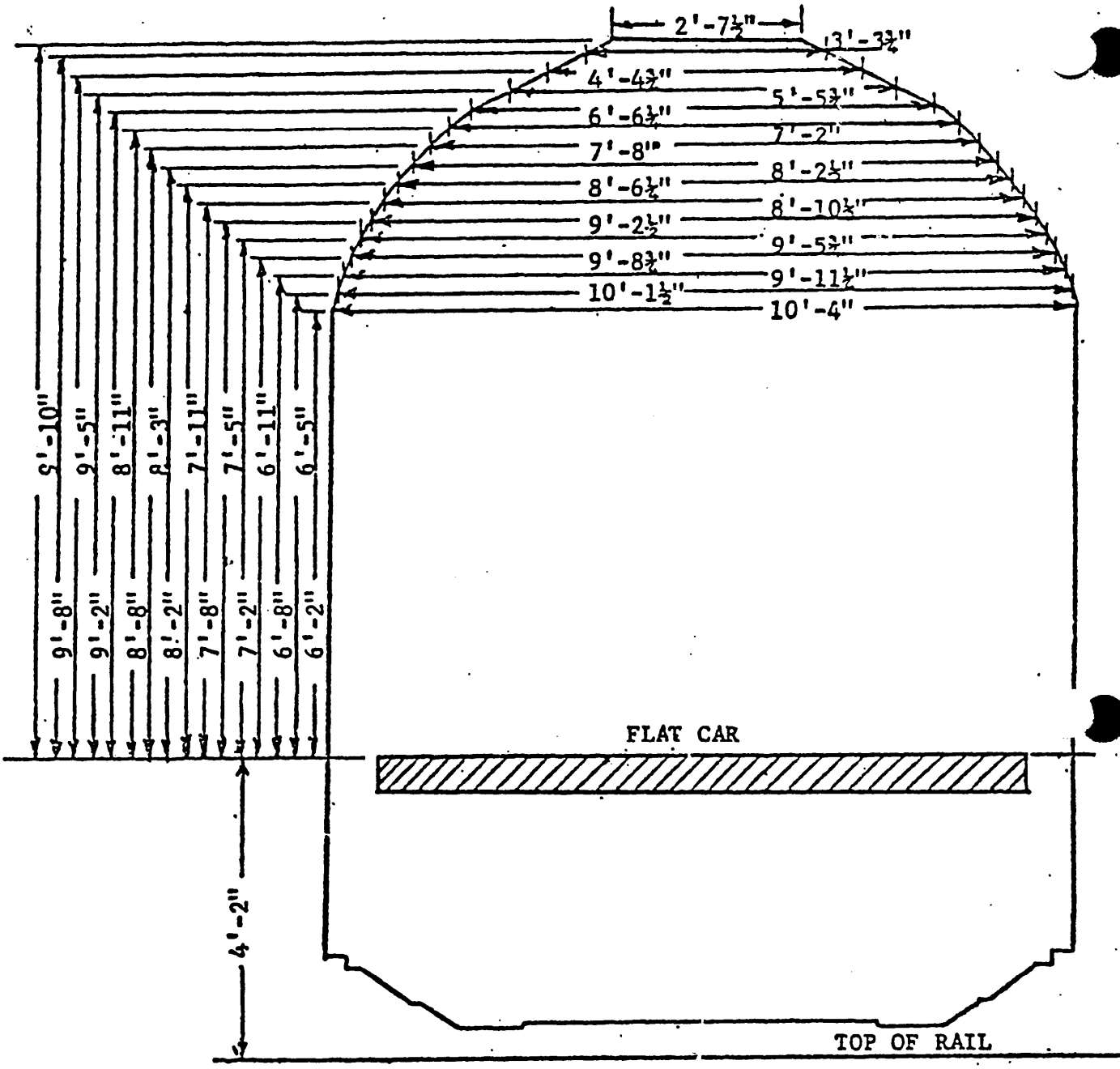


FIGURE 2: Item Length-32'-10"
BERNE INTERNATIONAL
(Clearance Diagram)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

4410-0039

DOCUMENT IDENTIFIER AND TITLE MIL-B-23170A BOILERS, STEAM, HIGH PRESSURE WATER TUBE, PACKAGED TYPE, (5,000 TO 50,000 POUNDS PER HOUR FIRED BY COAL, OIL,

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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